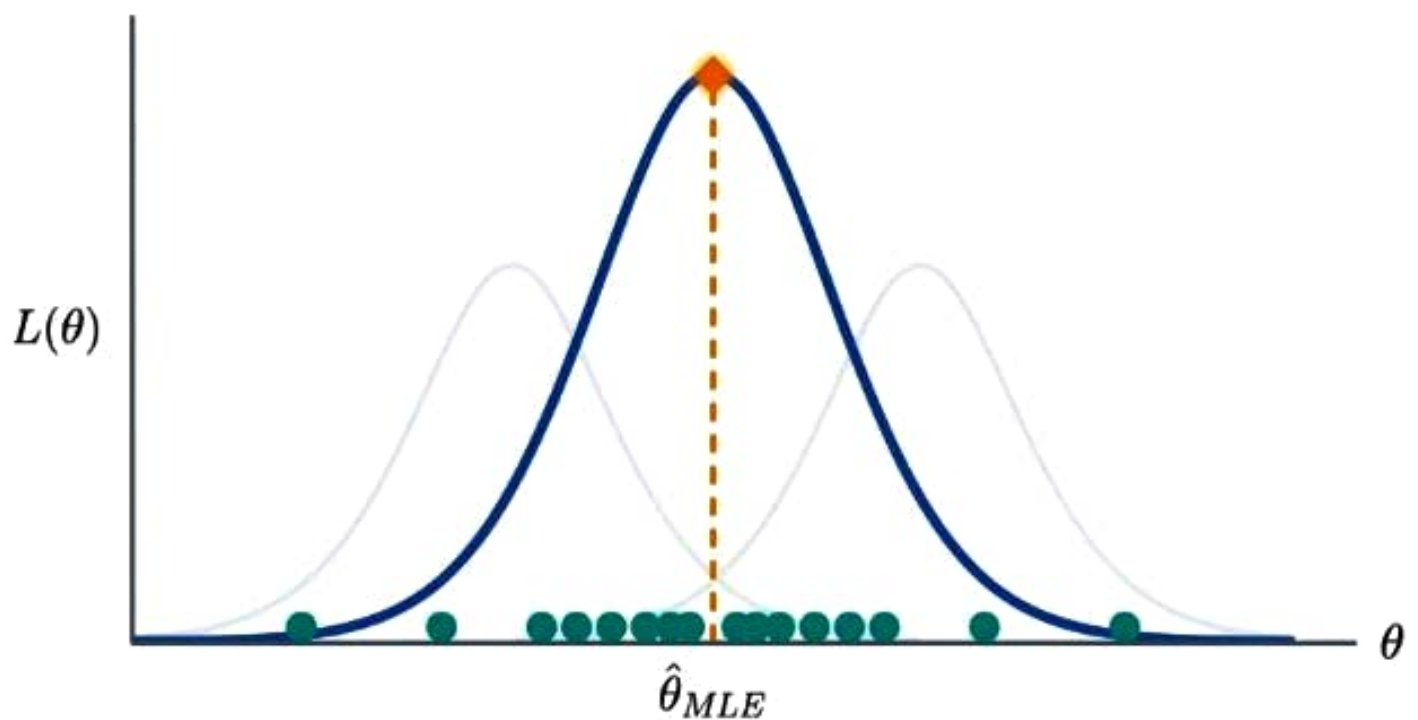




Maximum likelihood we try to maximize the gaussian which best fit our given data like in the given figure theta mle gives the best gaussian that fits the data

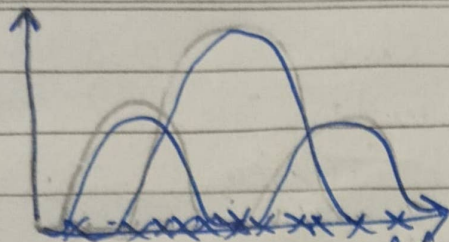
$$p(x_1, x_2, x_3, \dots | \theta) = \prod_{i=1}^N P(\mathbf{x}_i | \theta)$$

Where theta is [thetha1, thetha2, thetha3] and x is [xi, xz, xk,] our vector of k dimension and it represent single data x1 Latent variable , parameters definition

GMM → em algorithm

TOPIC

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$$p(x|z) = N(x_k, \Sigma_k, \mu_k)$$

$$L(\theta) = p\left(\frac{x}{\sigma}\right) = \prod_{x=1}^{x_2} \left(\sum_{k=1}^{K_2} p(x|z) p(z) \right)$$

likelihood

$$\ln L(\theta) = \ln \left(p\left(\frac{x}{\sigma}\right) \right) = \sum_{x=1}^{x_2} \ln \left(\sum_{k=1}^{K_2} p(x|z) p(z) \right)$$

not a closed form solution
hard to solve directly
Iteration

$$\frac{\partial S}{\partial \mu_k} = \pi_k \frac{\partial N(x_i | \mu_k, \Sigma_k)}{\partial \mu_k}$$

as all μ_k are independent
so only Normal distribution with mean μ_k will be left

* here z is a latent variable which means it is different from each x_i (different From Prior Distribution)

$\pi_k \rightarrow$ prior distribution

↳ it is same for each x and is different from responsibility

eg $x_1 \rightarrow z_1$ 0.3 $x_2 \rightarrow z_1$ 0.2 $x_3 \rightarrow z_1$ 0.4

z_2 0.4

z_3 0.3

$x_2 \rightarrow z_2$ 0.2 $x_3 \rightarrow z_2$ 0.3

$z_3 \rightarrow 0.4$

the overall as average contribution of element existing gaussian

this term is dependent on μ_k
we cannot directly solve it

$$\sum_j \pi_j N(x_n | \mu_j, \Sigma_j)$$

So we know $\gamma_k =$

Estep

↳ it is about finding the responsibility or the percentage of x present in each gaussian then we know all the value of x present in each gaussian based on old parameter

$$\gamma_{kz} p(z_k = k | x_n) = \frac{p(x|z) p(z)}{p(x)}$$

Mstep: Now we can update our parameters like we do in a single gaussian

$\mu_k \leftarrow \frac{\sum_{x \in X} \gamma_{kz} x}{\sum_{x \in X} \gamma_{kz}}$ But each gaussian has γ_{kz} of x so $\mu_k = \frac{\sum \gamma_{kz}(x)}{N}$

and similar for gaussian

$$\mu_k^{\text{new}} = \frac{\sum_{i=1}^N \gamma_{ki} x_i}{N_k}$$

$$\pi_k^{\text{new}} = \frac{N_k}{N}$$

$$\Sigma_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_{ki} (x_i - \mu_k^{\text{new}})(x_i - \mu_k^{\text{new}})^T$$

we will do this till

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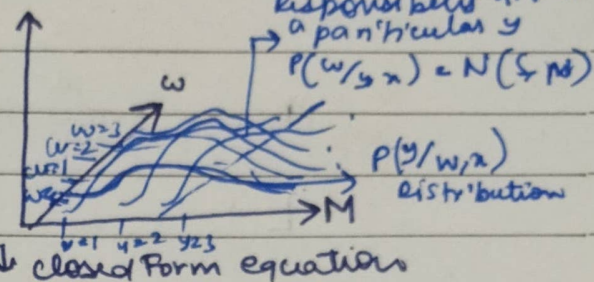
Vector → EM Algorithm

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$\omega \sim N(0, I)$ ^{Continuous latent variable}
 $M = m + T\omega + \text{error}$
 $\epsilon \sim N(0, \Sigma)$
 $\therefore M \sim N(m + T\omega, \Sigma)$

$P(M/\omega, T, m, \Sigma) \sim N(M/m + T\omega, \Sigma)$

$L(\theta) = L(\frac{m}{T}, m, \Sigma) = \prod_{m=1}^N \left(\int P(M/\omega) P(\omega) d\omega \right)$
 _{$m=0$ all the utterance.}



$\ln(L(\theta)) = -N/2 \ln |TT^T + \Sigma| - \frac{1}{2} \sum_{i=1}^N (M_i - m)^T (TT^T + \Sigma)^{-1} (M_i - m) + C$

However solving T, Σ which are interlinked is tough so we use EM Algorithm

$P(\omega_i/M_i) \propto P(\frac{M_i}{\omega_i}) P(\omega_i)$ (Bayes theorem)

Plotted

Gaussian

Prior Gaussian

$P(\omega_i/M_i) \rightarrow$ follows Gaussian too

where $N(S, T^T \Sigma^{-1} M, \frac{1}{2} (I + T^T \Sigma^{-1} T)^{-1})$

This is the distribution of responsibility for a given M_i [how M_i is decided]

Here EM step is finding $E[x_i]$ and $E[\omega_i \omega_i^T]$ as $Q(T, \Sigma) \rightarrow$ depends on that

$Q(T, \Sigma) = \sum_{i=1}^N \left[-\frac{1}{2} \ln |\Sigma| - \frac{1}{2} (M_i^T \Sigma^{-1} M_i - 2 M_i^T \Sigma^{-1} E[\omega_i] + \text{Tr}(T^T \Sigma^{-1} E[\omega_i \omega_i^T T])) \right]$
_{Trace}

$\frac{dQ}{dt} = 0 \rightarrow T_{\text{new}} = \left(\sum_{i=1}^N \tilde{M}_i E[\omega_i] E[\omega_i]^T \right) \left(\sum_{i=1}^N E[\omega_i \omega_i^T] \right)^{-1}$

$\Sigma_{\text{new}} = \frac{1}{N} \sum_{i=1}^N \left(M_i M_i^T - 2 T_{\text{new}} E[\omega_i] M_i^T + T_{\text{new}} E[\omega_i \omega_i^T] T_{\text{new}} \right)$

$\ln P(x/\theta) = \ln \left(\sum_z P(x, z/\theta) \right)$

to maximize we bring an arbitrary distribution

$\ln P(x/\theta) = \ln \left(\sum_z q(z) \frac{P(x, z/\theta)}{q(z)} \right)$

$\ln P(x/\theta) = \ln \left(\sum_z q(z) \frac{P(x, z/\theta)}{q(z)} \right)$

Jensen's Inequality $\ln E[Y] \geq E[\ln Y]$

$\ln P(x/\theta) \geq \sum_z q(z) \ln P(x, z/\theta)$

$L(q, \theta) = \sum_z q(z) \ln P(x, z/\theta) = \sum_z q(z) \ln \dots$

KL divergence

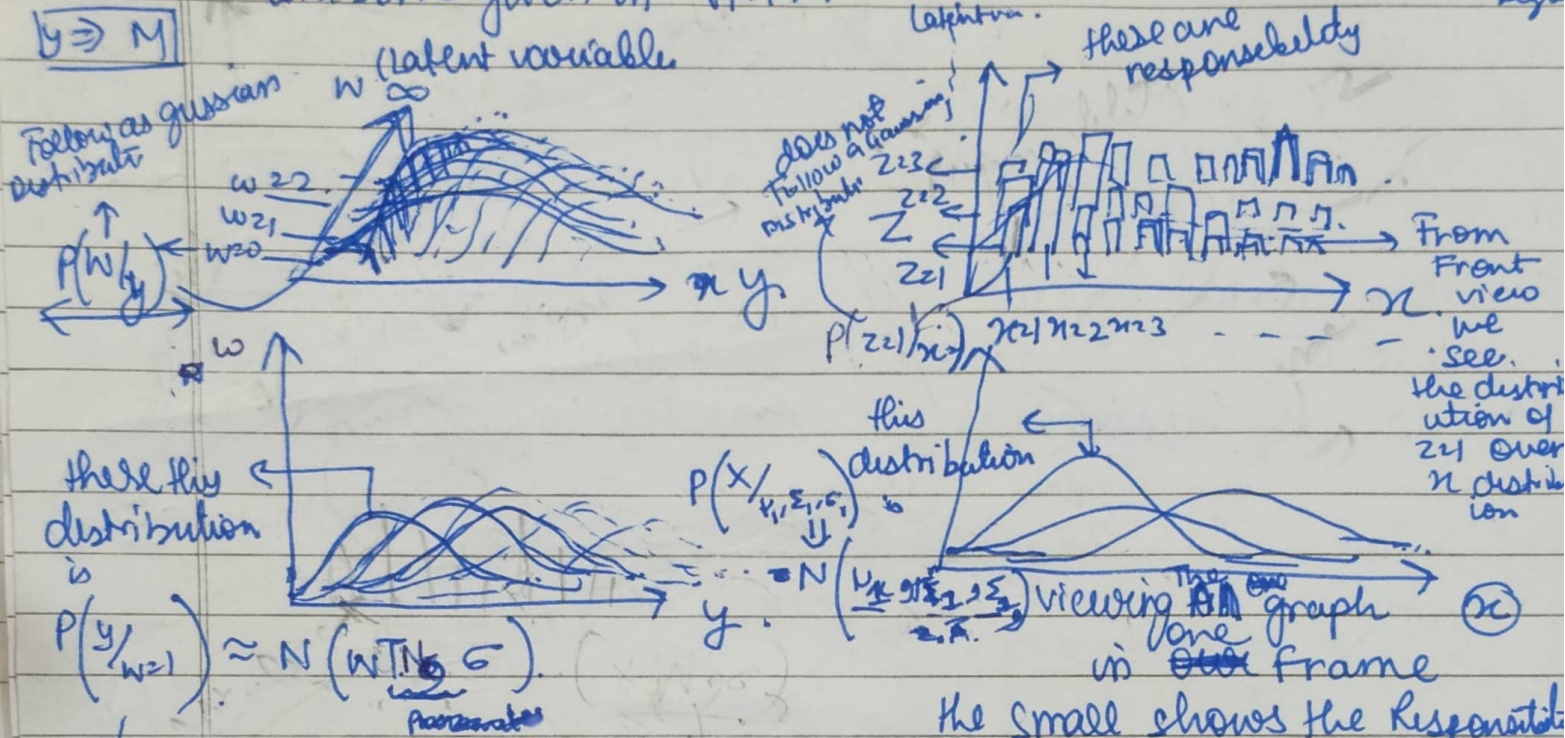
KL Diverge 20 putting $q(z) = P(z/x, \theta, \text{old})$ Q function

$L(q, \theta) = \sum_z P(z/x, \theta, \text{old}) \ln P(x, z/\theta)$

to maximize $\sum_z P(z/x, \theta, \text{old}) \ln P(x, z/\theta)$

$\frac{dL(q, \theta)}{d\theta} = \frac{dQ}{d\theta} = 0$ only old param

→ Our Answered how are EM Algorithm and Latent variable connected in vector and GMM??
we assume Infinite many Gaussian in Vector/Bayesian and some given in GMM



here each distribution has different w , however same μ and σ and N_c is a given But different for each z

So there are some similarities between GMM and Σ vector Implementation

But major difference is that the Gaussians used in Σ vector are interconnected and do not have independent parameters

to use EM we need latent variable

where $m_n = \beta^T S_n^T X^T Y$
 $S_n^T = \alpha I + B^T X^T X$

